

Research in the Wild

What it takes to push academic work to the (open) frontier

Scott Geng x Nathan Lambert, July 2026

Part 1: The Research

Preference Tuning Basics (2022-2023)

- Dataset = prompt x , chosen y_c , rejected y_r
- RLHF objective (InstructGPT, Ouyang et al. 2022)

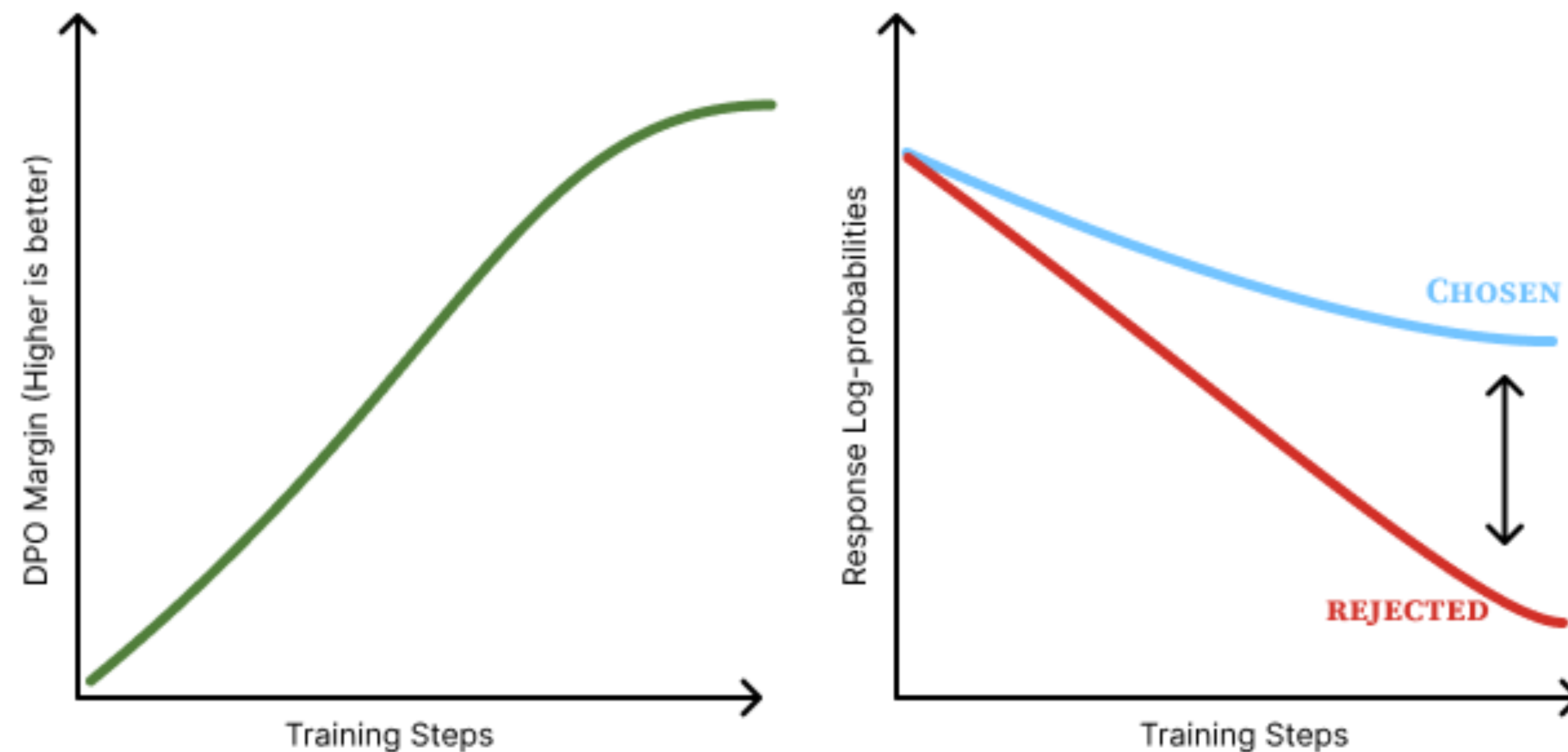
$$\max_{\pi} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x)} [r(x, y)] - \beta \mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x))]$$

- Direct Preference Optimization (Rafailov et al. 2023)

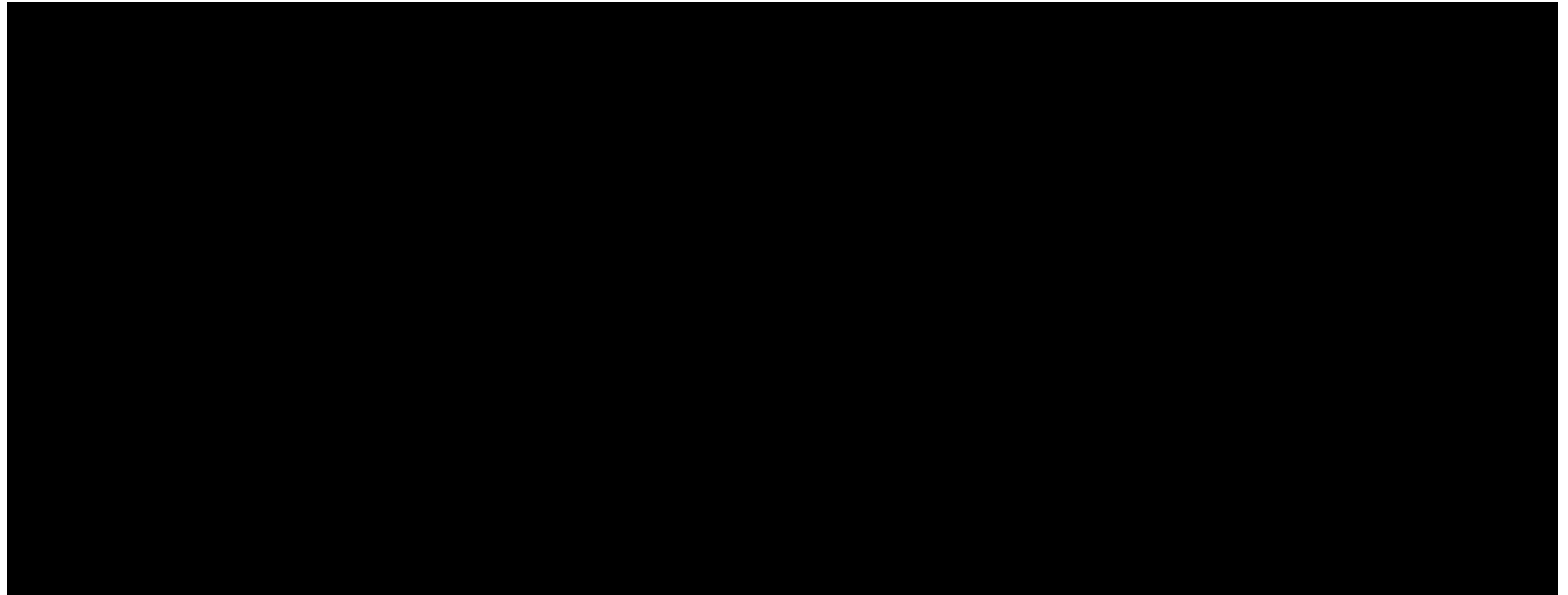
$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_c, y_r) \sim \mathcal{D}} \left[\log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(y_c | x)}{\pi_{\text{ref}}(y_c | x)} - \log \frac{\pi_{\theta}(y_r | x)}{\pi_{\text{ref}}(y_r | x)} \right] \right) \right]$$

Algorithm Zoo (2023 - 2024)

- SimPO, ORPO, IPO, ODPO, REBEL, APO...
- KL resets, length normalization, NLL regularization...

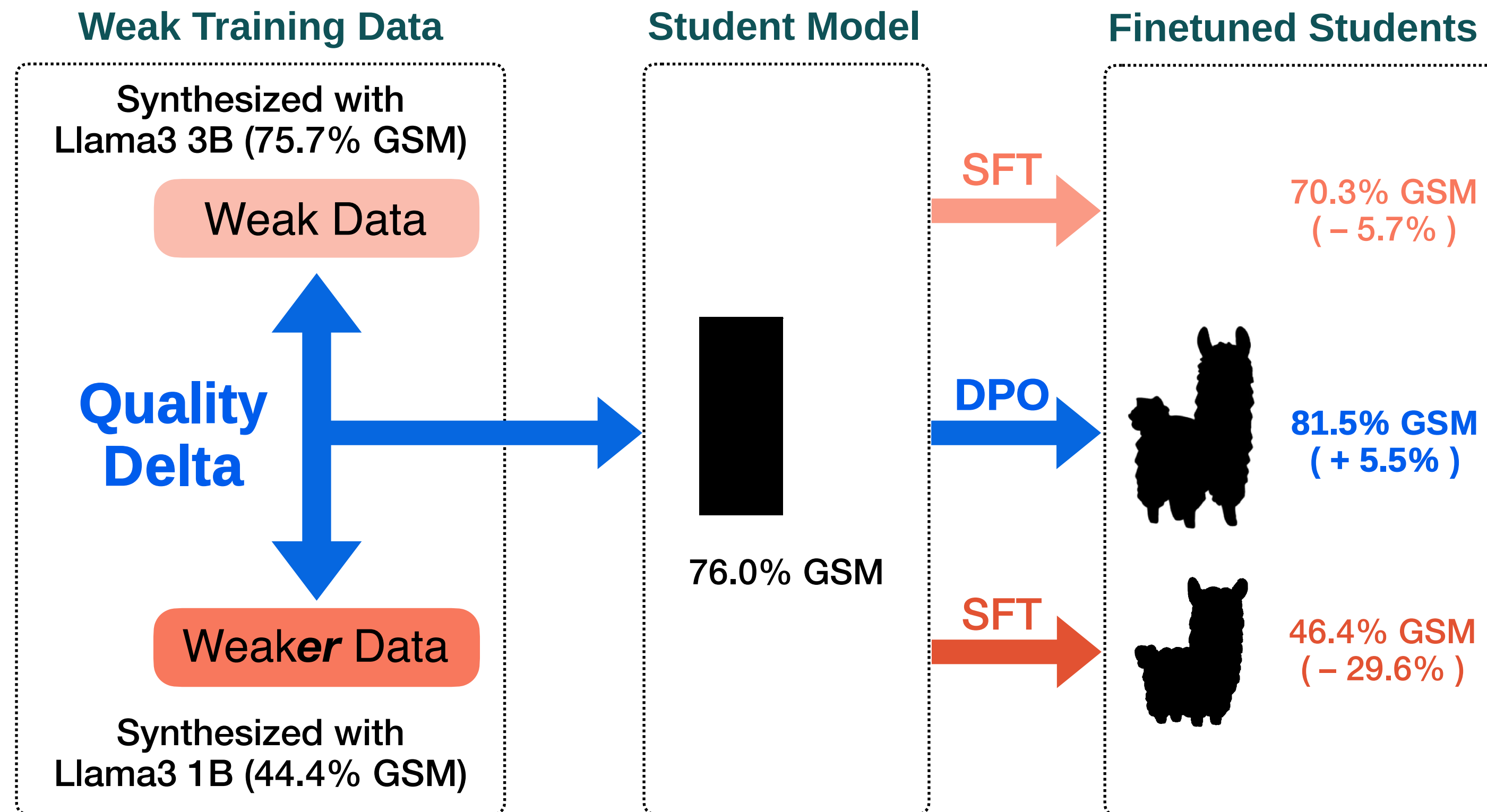


Tulu 3 / Ultrafeedback (2023 - 2024)



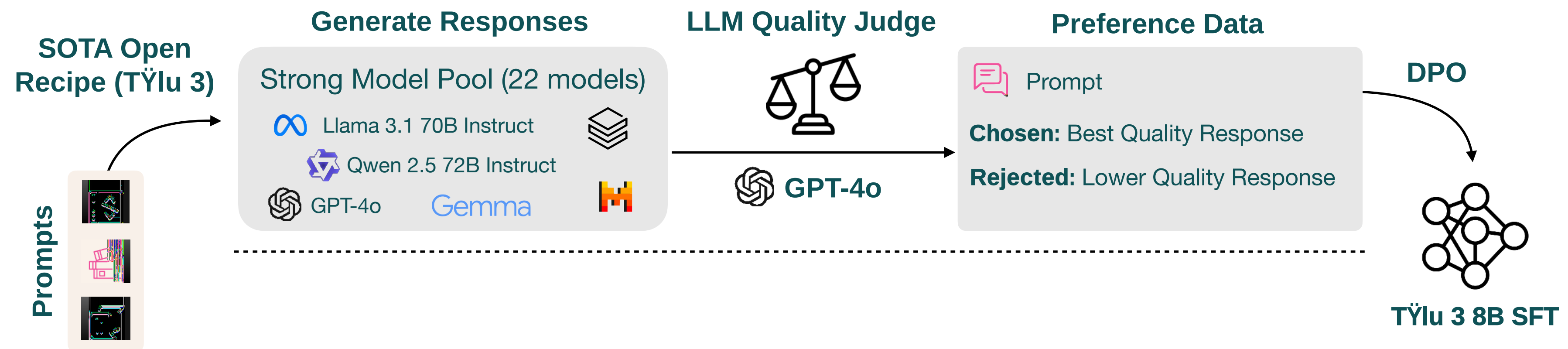
The Delta Learning Hypothesis (Early 2025)

What matters when learning with paired data?



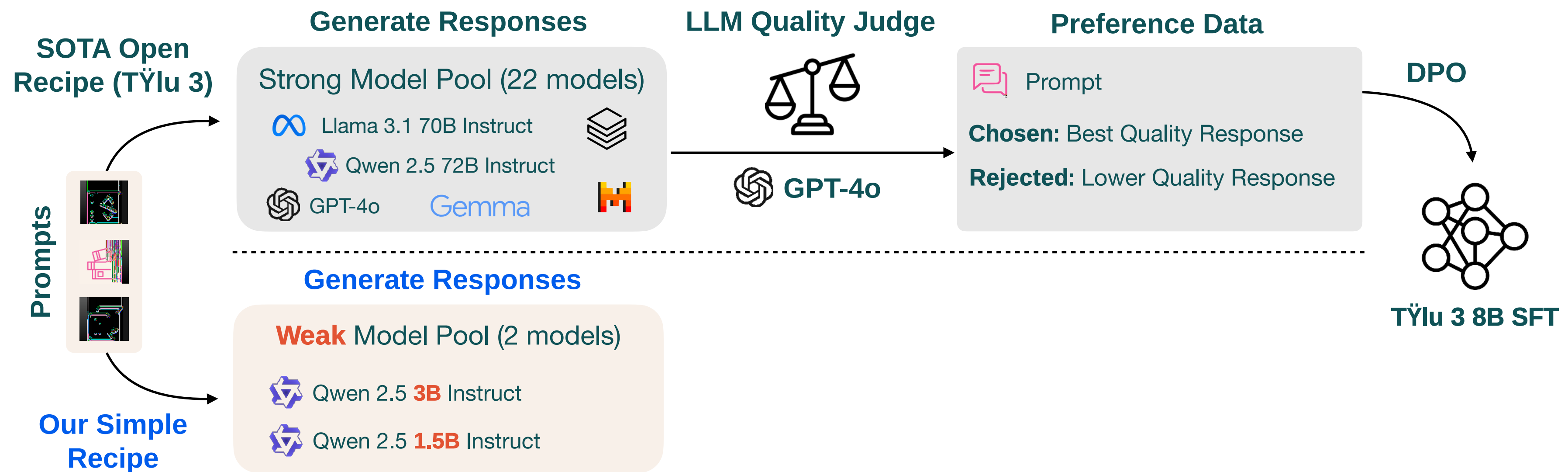
The Delta Learning Hypothesis

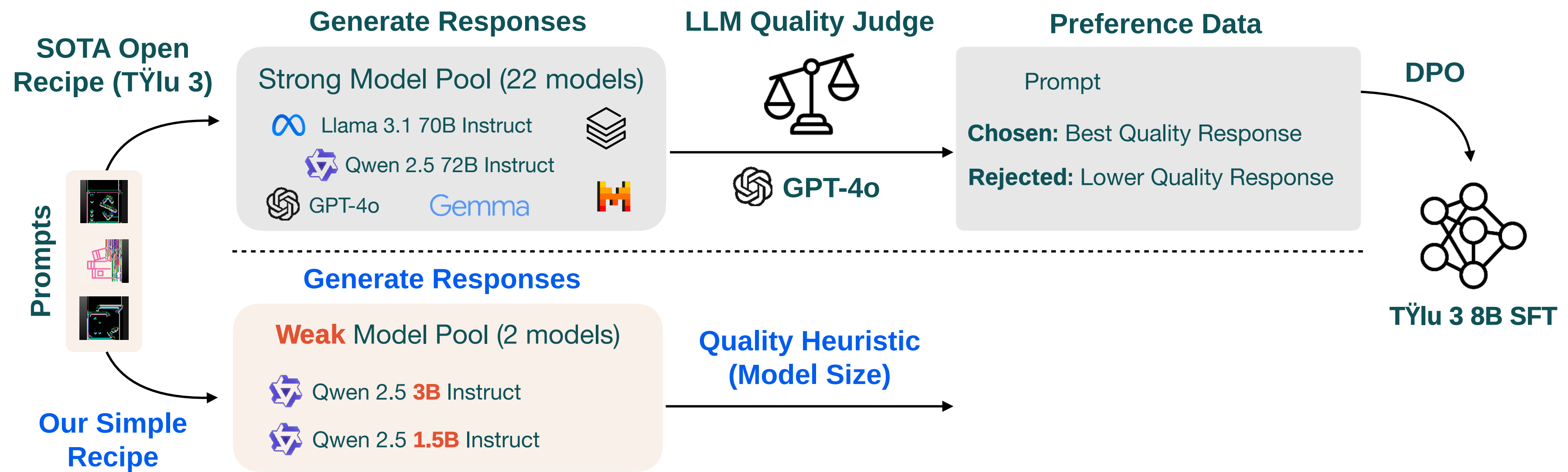
Post-training with Ultrafeedback



The Delta Learning Hypothesis

Post-training with ~~Ultrafeedback~~ only weak data

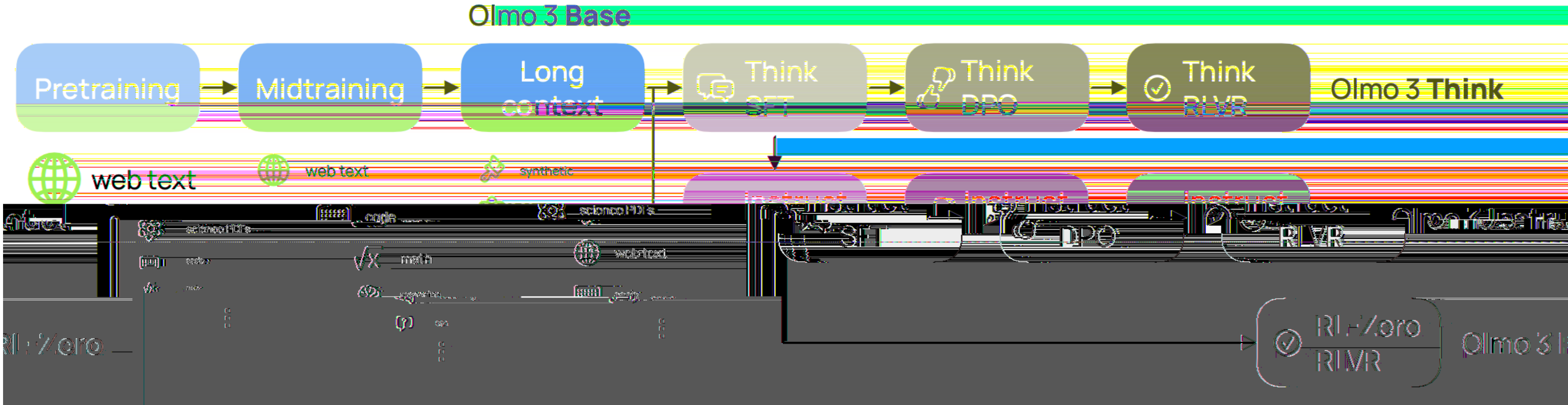




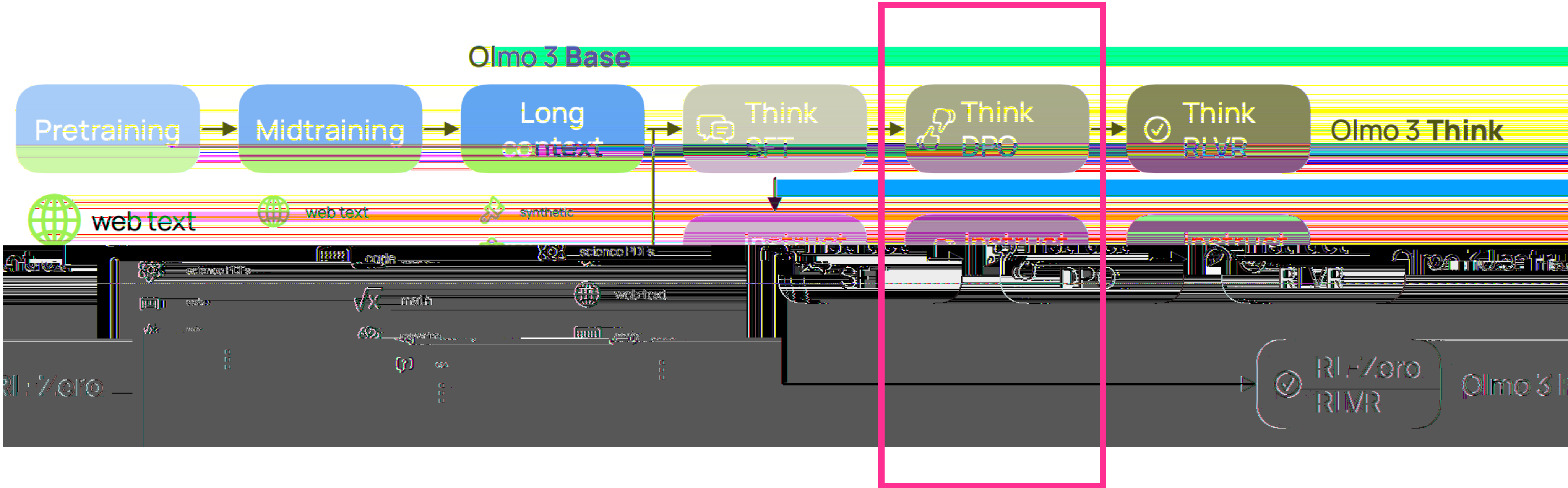
Stronger Model Fit in 2024 (77%) vs. 2023 (70%)

Part 2: The Wild

Olmo 3 (Dec 2025)

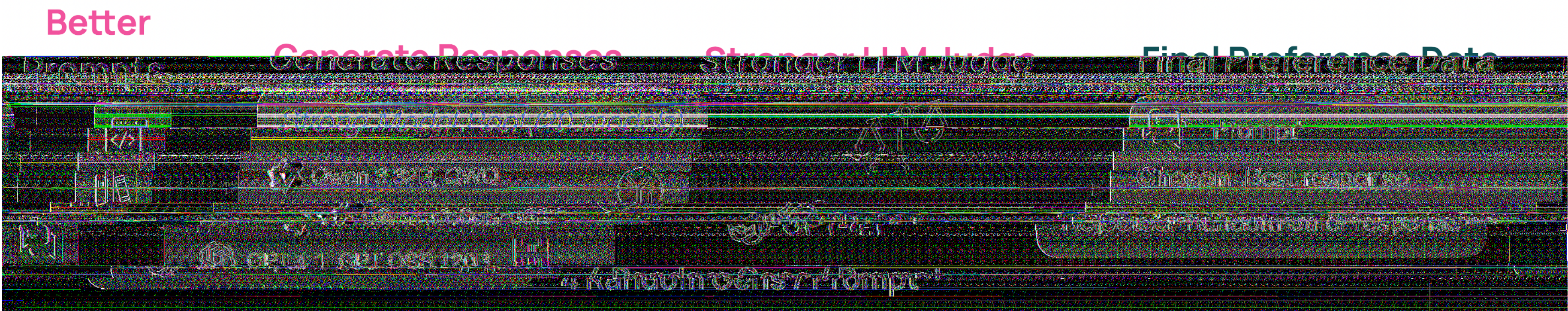


Olmo 3 (Dec 2025)



Olmo 3 (Dec 2025)

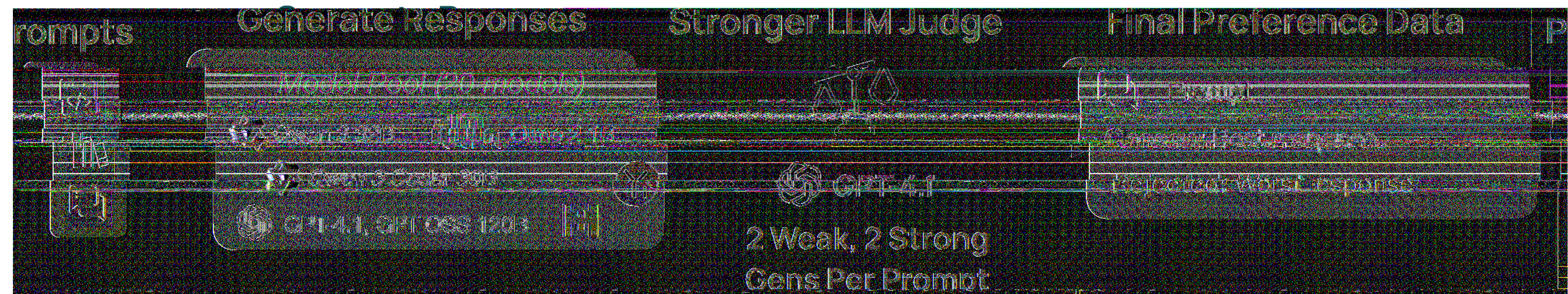
Where we start



Olmo 3 (Dec 2025)

Where we end - how to fix DPO

Better

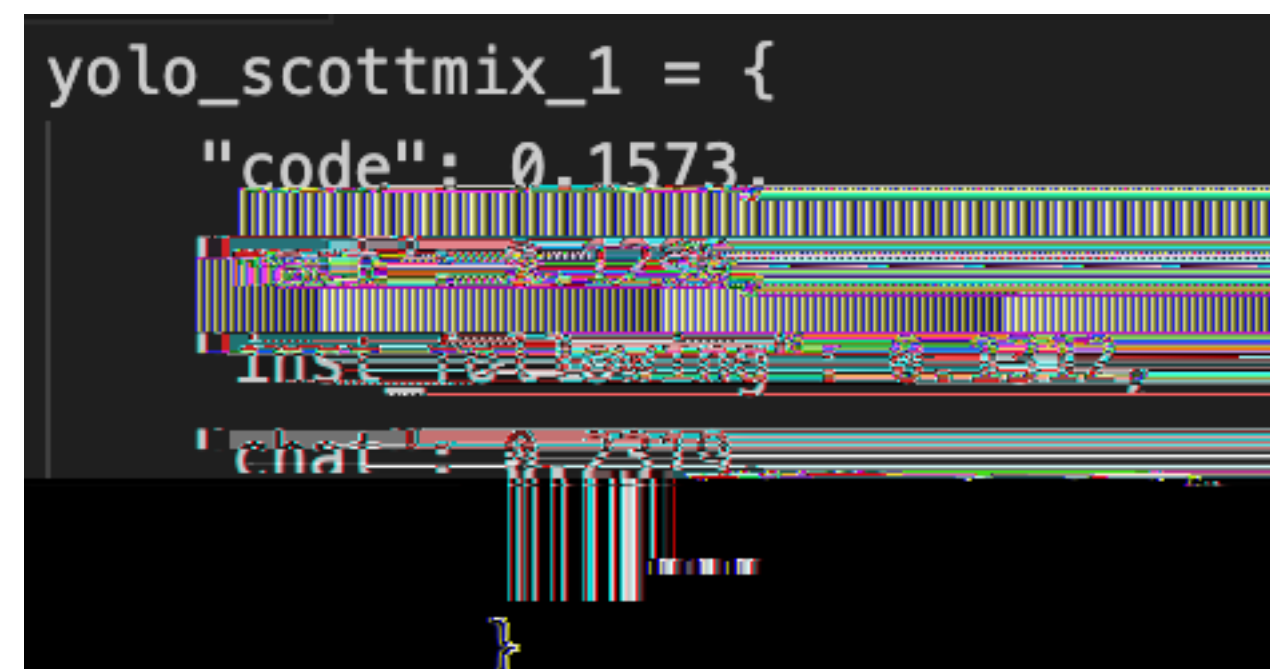


Force weak models in the judge pool, pick worst response, and use delta learning heuristic pairs

SmolLM3 converged to a similar preference data setup (Bakouch et al. 2025)

Challenge 1: Unstudied data details

- “My name is ChatGPT / Deepseek / Qwen”
- Better on hallucination evals, but more hallucination vibes?
 - Came from length bias
- Prompt mixing = wtf

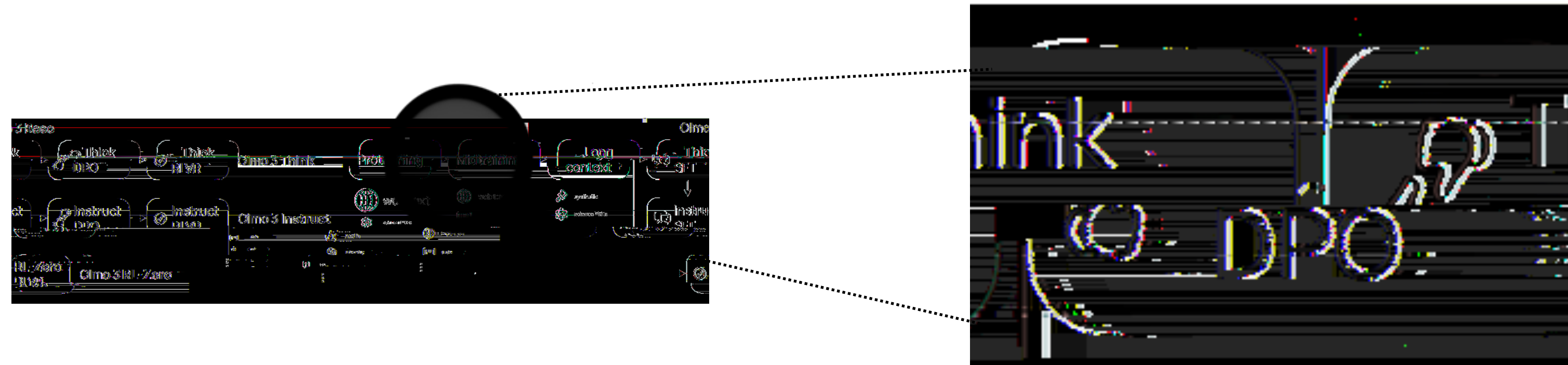


```
yolo_scottmix_1 = {  
  "code": 0.1573,  
  "inst": "  
  "chat": "  
}
```

Challenge 1: Unstudied data details

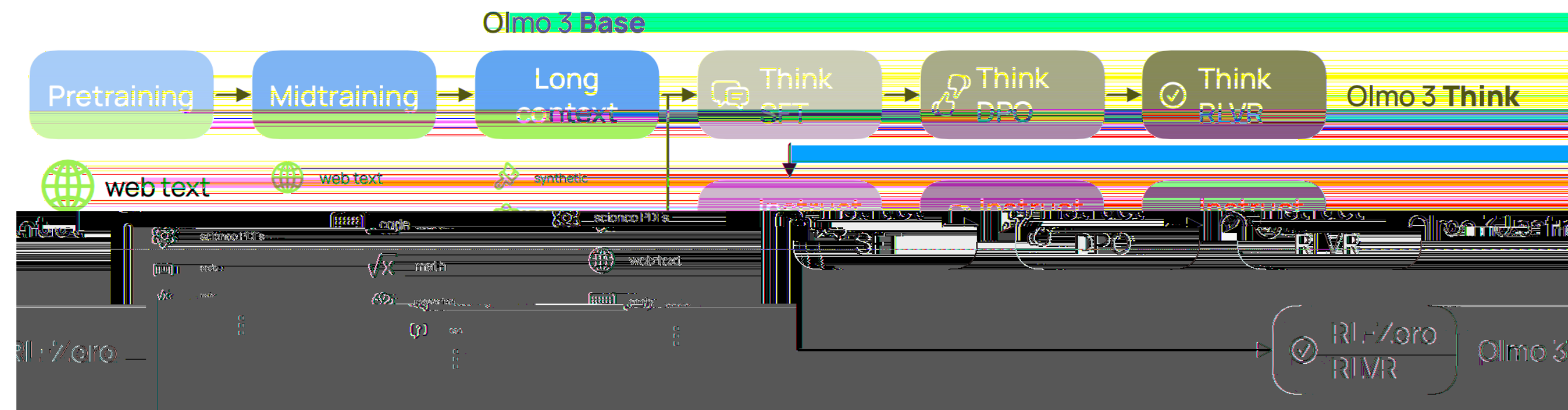
- “My name is ChatGPT”
- Better on hallucinations?
 - Came from length
- Prompt mixing = w

Challenge 2: Organization / method sequencing



- In research, preference tuning is a single, standalone unit of study ●

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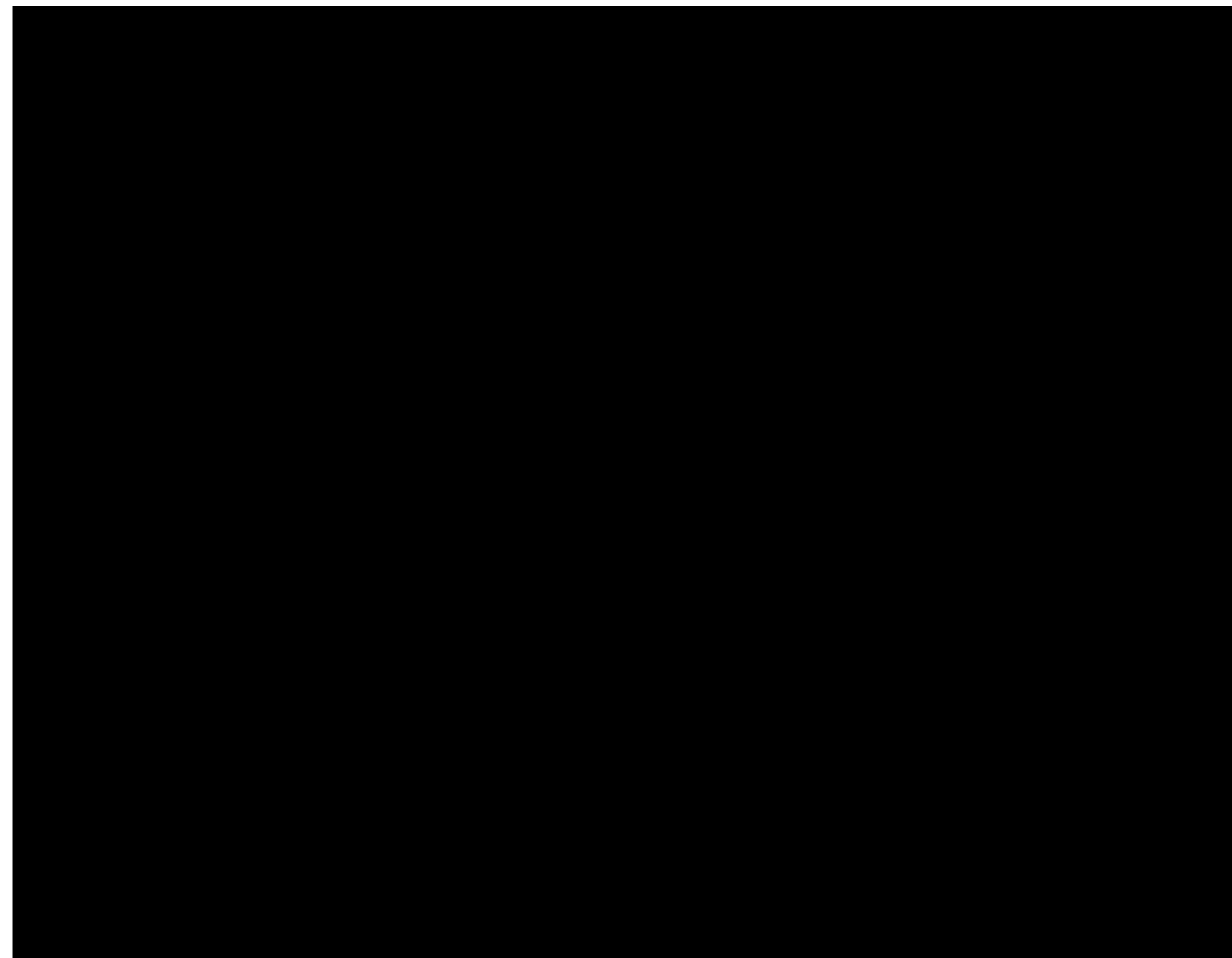


- In research, preference tuning is a single, standalone unit of study ●
- In practice, one part of a larger pipeline / organization, all moving in parallel ●

SFT A < SFT B, but SFT A + DPO >> SFT B + DPO

Challenge 3: ???

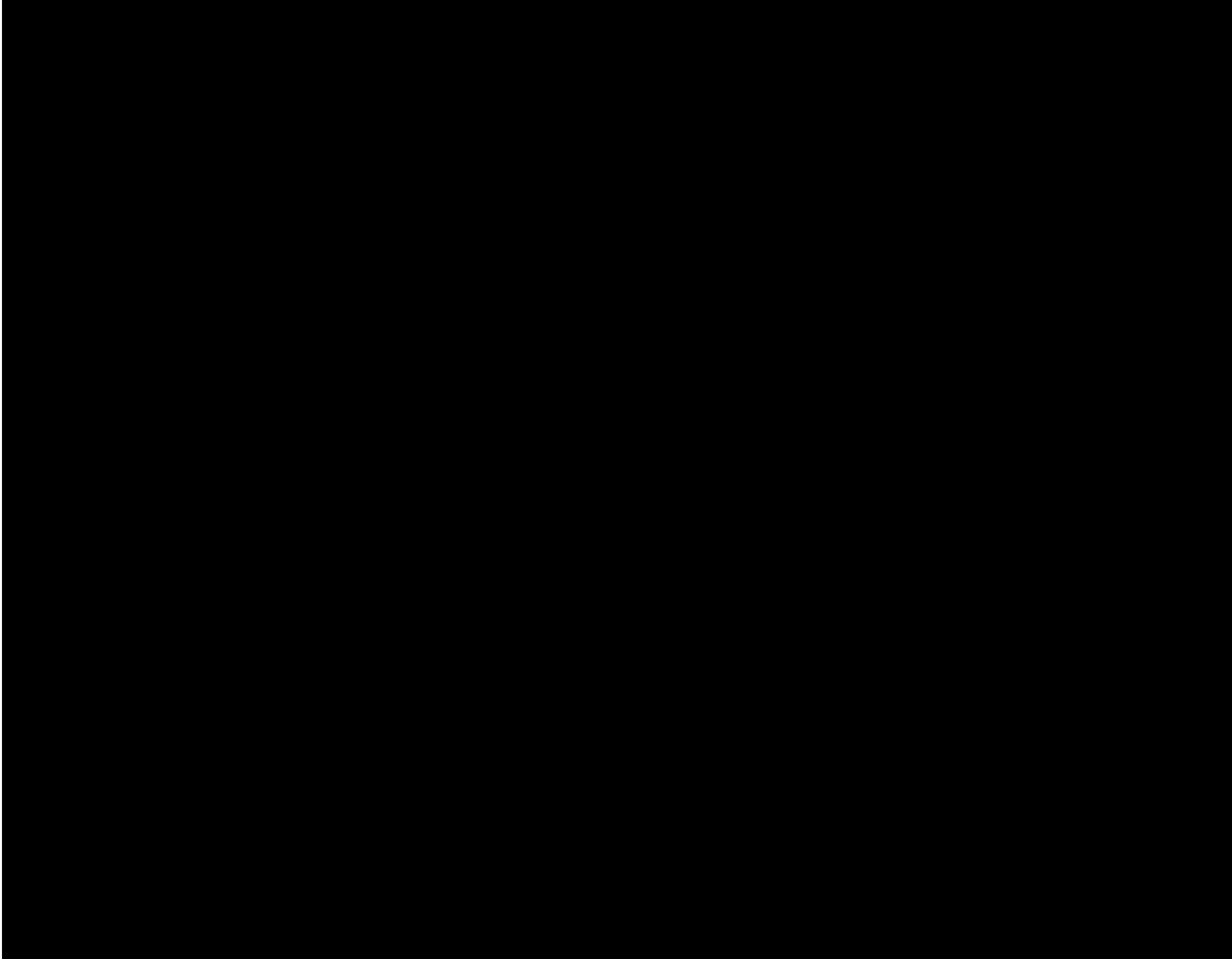
train_loss



At larger scale, every random problem that can hit will

Challenge 3: ???

train_loss



At larger scale, every random problem that can hit will

Fix: NCCL vars +
DeepSpeed sharding
config + forward pass
details + caffeine

The future?